Fluid Flow versus Euler's Equations versus OpenAI

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Abstract

Three things are to be kept apart: a fluid flow, which is a phenomenon; Euler's equations, which are a model of it; and a proof about the equations, which is a statement about the model. We state the equations, note that they are parameter free, and observe that the mathematical question asked of them — existence, regularity, uniqueness — has not been answered by analysis in symbolic form, because the solutions turn out to have a complicated nature which symbolic analysis has no grip on. Solutions in digital form over a mesh can be computed by a model we name *Adaptive Euler* — a stabilised finite element discretisation together with duality-based output error control and adaptive refinement — and they are turbulent, fluctuating over a range of scales, with well determined mean values such as drag and lift which agree with measurement and arrive with a computed bound on their error. We then ask what a recent symbolic proof of finite-time blowup establishes, and conclude that it is a statement about the model which does not reach either of the other two levels: it exhibits no individual case, so nothing can be inserted into the equations, inspected, or measured.

1 The equations

Let $\Omega \subset \mathbb{R}^3$ be a domain with boundary Γ, let $x = (x_1, x_2, x_3)$ be a Euclidean coordinate in Ω and t time. An inviscid incompressible fluid of unit density filling Ω is described by a velocity $u(x,t) \in \mathbb{R}^3$ and a pressure $p(x,t)$ satisfying, for $0 < t \leq T$,

$$\dot{u} + (u \cdot \nabla)u + \nabla p = f \qquad \text{in } \Omega, \tag{1}$$

$$\nabla \cdot u = 0 \qquad \text{in } \Omega, \tag{2}$$

together with the *slip* boundary condition

$$u \cdot n = 0 \qquad \text{on } \Gamma, \tag{3}$$

n the outward normal, and an initial velocity $u(\cdot, 0) = u^0$ in Ω.

Equation (1) is Newton's second law. For unit density the mass of a fluid particle is its volume, so force equals acceleration directly: the left-hand side $\dot{u} + (u \cdot \nabla)u$ is the acceleration of a particle following the flow, and the right-hand side is the force acting on it — the pressure gradient $-\nabla p$, which is the force the surrounding fluid exerts, together with any body force f. Equation (2) states incompressibility: the volume of a fluid particle does not change, so the velocity field is divergence free, and the pressure is whatever field enforces this. Condition (3) states that the fluid does not pass through the wall, and nothing more. It is the only boundary condition the equations admit: the normal component is fixed by impermeability, nothing

in (1)–(2) determines the tangential velocity at the wall, and prescribing it as well would overdetermine the problem. The no-slip condition belongs to Navier–Stokes, where a second-order viscous term carries the extra condition; here there is no viscosity, and therefore no boundary layer either.

What 'inviscid' means. Not that the viscosity is zero. Air and water have viscosity, and no flow of either is literally inviscid. What is meant is that the Reynolds number is large — 10^6 to 10^8 in the flows of aerodynamics and ship hydrodynamics — so that viscous forces are negligible beside inertial ones everywhere except in layers which are thin and, as it turns out, do not determine the forces on the body. Euler's equations are the model of that limit, and the limit is where the flows of practical interest live. The question is never whether a real fluid has viscosity. It is whether the quantities one wants depend on how much.

The model is parameter free. There is no constant in (1)–(3) to be measured, fitted or chosen. What enters as data is the *geometry* — the domain Ω, which is to say the shape of the body the fluid flows past — the *forcing* f, and the initial state u^0. Nothing else. Whatever the equations predict is therefore a consequence of the geometry and the forcing alone, and any disagreement with observation is a fault of the model and cannot be repaired by adjustment.

2 The mathematical question

The question mathematics asks of a system such as (1)–(3) is about the *properties of its solutions*. Does a solution exist for the given data? Is it regular — does it stay smooth, or do derivatives become unbounded? Is it unique, so that the data determine the flow?

These are the right questions to ask and they have not been answered in symbolic form. The reason is not that the problem has been neglected; it is that the solutions turn out to have a complicated nature. A flow at a Reynolds number of any interest is turbulent: it fluctuates on a range of scales from the size of the body down to scales many orders smaller, with no symmetry, no similarity variable and no closed form. Symbolic analysis works by rewriting, and rewriting needs something to grip. Against an object defined by carrying structure at every scale with no organising principle to exploit, there is nothing for it to take hold of, and this is not a temporary state of the technique but a description of what turbulence is.

What symbolic analysis has produced instead are statements about objects which do possess exploitable structure — self-similar profiles, symmetric configurations, families built by iteration — and conditional bounds of the form: if the velocity is bounded by A, then the derivatives are bounded by an expression exponential in A iterated several times. As theorems these are real. As statements about a fluid they are empty, since a bound of size $\exp\exp\exp A$ is not one any flow can be said to respect.

3 Adaptive Euler: solutions in digital form

The same equations treated in digital form behave quite differently. We name the computational model **Adaptive Euler**. It has three parts and the third is not an afterthought:

1. the equations (1)–(3), discretised by a stabilised finite element method over a mesh [4], the stabilisation acting as a vanishing artificial viscosity so that what is computed is a *viscosity solution* of Euler, the limit taken as the mesh is refined;

2. *duality-based output error control*: for each output of interest — a drag, a lift — the error is bounded by the residual of the computed solution weighted by the solution of an adjoint problem, which measures how much the residual matters for that output;

3. *adaptivity*: the mesh is refined where the weighted residual says the error lives, and not uniformly, until the bound on the output is below what is asked.

The error control is part of the model rather than an assessment of it. What Adaptive Euler returns is not a flow but a *pair*: an output and a bound on how wrong that output may be, computed together. A model which returned the first without the second would be a different model and a much weaker one, since nothing could then be concluded from it.

The viscosity introduced, and why its size does not matter. A computation at finite resolution cannot have zero dissipation, so Adaptive Euler necessarily introduces a viscosity. This is the obvious objection to it — that what is being solved is not Euler's equations but some nearby viscous system — and the answer is a computational fact rather than an argument.

The mean-value outputs do not depend on the absolute size of the viscosity introduced. Refine the mesh: the artificial viscosity falls, the resolved scales multiply, the flow field changes in every detail — and the drag does not move. The dissipation rate does not fall with the viscosity either; it settles at a value fixed by the geometry and the free stream. That independence is the dissipation anomaly in computational form, the phenomenon Onsager accounted for [9] and Taylor measured, and it is the reason the limit is meaningful at all.

It also places the computation and the fluid in the *same* position rather than in different ones. Air has a small viscosity whose absolute size does not determine the drag of a wing; Adaptive Euler has a small viscosity whose absolute size does not determine the computed drag. Neither is inviscid and neither needs to be. What both exhibit is a flow in which the viscosity sets the smallest scale present and nothing that is asked about.

What Adaptive Euler produces has the following properties, and none of them was put in.

It is turbulent. The computed flow fluctuates, in time and across a range of scales widening as the mesh is refined. It is not a steady solution, not a symmetric one, and not one that could have been guessed.

Its mean values are determined. The flow field itself is not reproducible in detail — and could not be, since for weak solutions of Euler uniqueness is false. But mean-value outputs are: drag, lift, the position of the separation line. These converge under refinement, and each arrives with the bound of part 2 above, so the answer carries a statement of how wrong it may be.

They agree with measurement. Drag within a few per cent, the separation line where experiment puts it, the lift curve and the stall, the drag crisis [7]. These are quantities measured

by other people at facilities with no interest in the computation, and in several cases submitted before the measurements were released.

And it is a solution of Euler. That the limit satisfies (1)–(3) weakly is not an identification offered because the numbers look right. The stabilisation of the method is a weighted least-squares control of the residual; the discrete energy estimate bounds that residual in a weighted norm; testing against a smooth function sends it to zero in the sense the weak formulation requires. The limit is a weak solution, as a theorem.

So the questions of Section 2 do have answers in digital form, though not the answers the symbolic form was looking for. Existence: by construction and exhibition. Regularity: none — the velocity stays bounded while its gradients do not, the solution being Hölder continuous with exponent about $1/3$ — the exponent below which Onsager observed that energy conservation can fail [9] — and dissipating energy at a rate independent of the vanishing viscosity. Uniqueness: false for the flow field, true for the mean values, which is the form in which the question has an affirmative answer and the form in which anybody wants one.

4 What the symbolic proof establishes

Against this stands a recent proof, produced by an AI system and formalised in Lean, that there exists smooth, compactly supported, divergence-free initial data on $\mathbb{R}^3$ whose smooth Euler solution has finite maximal lifespan: the velocity gradient becomes unbounded, and the Beale–Kato–Majda [8] vorticity integral diverges [1]. Comparable results were being prepared independently and by hand [2]. We take the proof to be correct, and nothing below depends on whether it is.

It is a statement about the *model*, and it is a real one: a question open for decades is settled. Three observations place it.

The velocity does not become infinite. What diverges is the gradient. That is the same signature as the computed turbulent solution — gradients unbounded, velocity bounded — so the two accounts agree that smooth solutions of (1)–(3) break down in finite time, which is what we reported computationally in [3]. The disagreement lies elsewhere.

It conserves energy, and therefore cannot be a fluid. With smooth data the solution is classical up to the blowup time, and a classical solution of (1)–(2) with decaying data satisfies

$$\frac{d}{dt}\int \tfrac{1}{2}|u|^2 = 0,$$

by an integration by parts. The construction therefore dissipates *nothing* over the whole of its existence. The singularity is not a mechanism by which energy is lost but one by which it is piled up: at the singular time all of it is still present, gathered into a vanishing region. A fluid loses energy continuously from the moment it meets a body, and the loss is the drag.

It exhibits no individual case. The datum is the limit of an iteration over stages and has no closed form. Nothing can be plotted, nothing measured. Nor can either check on a *result*

be applied: there is no symbolic expression to insert into (1)–(2) and reduce, and no numerical field either, since no finite stage of the iteration blows up — the property belongs to the limit. What exists is the existence.

5 Three levels, and what reaches which

The title names three things and the point of the paper is that they are three.

A fluid flow is a phenomenon. It has a drag which can be measured.

Euler's equations are a model of it, parameter free, with geometry and forcing as data. They are answerable to the phenomenon: if the drag they give is wrong, nothing can be adjusted to repair it.

A proof about the equations is a statement about the model. It is answerable to the model alone.

The digital solution reaches all three levels. It is a solution of the equations, by the residual control; it is an object that can be inspected, and inspecting such objects is where the resolution of d'Alembert's paradox [5] and the account of flight [6] came from; and its mean values can be set beside a measurement, where they survive.

The symbolic proof reaches the first of the three and stops. That is not a criticism of it as mathematics, where it is an advance, and existence without an exhibited instance is ordinary and accepted practice — the disproof of the Mertens conjecture [10] established that a counterexample exists and exhibited none. It is a statement about what the result can be offered as. A theorem about a model becomes a statement about fluids only by way of an instance, and there is none here.

6 Summary

The equations (1)–(3) are parameter free and have only geometry and forcing as data. The mathematical questions asked of them — existence, regularity, uniqueness — have gone unanswered in symbolic form because turbulent solutions have no structure for symbolic analysis to exploit, this being what turbulence is rather than a gap in technique. In digital form, by Adaptive Euler, the same questions are answered, in altered shape: solutions exist by exhibition, are not regular, are not unique as fields, and are unique in the mean values anybody wants, with computed error bounds and agreement with measurement.

A symbolic proof of finite-time blowup settles a long-open question about the model. Its solution conserves energy exactly and so dissipates nothing, whereas a fluid dissipates and the dissipation is the drag; and it exhibits no individual case, so nothing can be inserted into the equations, inspected or measured. It is an exact statement about an object with no observables, set beside approximate statements about observables which agree with experiment. Each has what the other lacks, and only one of them is answerable to a fluid.

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